

Multi-Environment TinyML Calibration of a Low-Cost Pan-Tilt Time-of-Flight LiDAR for Indoor 3D Reconstruction

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Abstract—Single-point time-of-flight (ToF) sensors provide a low-cost alternative to mechanically scanned LiDAR for indoor mapping, but their ranging accuracy is sensitive to target reflectivity, ambient illumination, multipath interference, and scan geometry. This paper presents a low-cost pan-tilt ranging platform built around a VL53L1X ToF sensor and an ESP32 microcontroller, and evaluates machine-learning-based residual calibration across two substantially different indoor acquisition campaigns. The first campaign was conducted in a relatively controlled, low-ambient-light room, whereas the second introduced stronger LED illumination, painted concrete surfaces, cardboard obstacles, and translucent objects. Cross-environment transfer experiments show that a model trained in one environment does not reliably transfer to the other: XGBoost produced mean absolute errors (MAEs) of 947.15 mm when trained on the controlled environment and tested on the diverse environment, and 1049.09 mm in the reverse direction. In contrast, pooled training across both environments reduced the combined-dataset MAE from 149.84 mm for the uncalibrated sensor to 22.41 mm with XGBoost. For embedded deployment, a single decision tree with maximum depth 12 achieved 23.42 mm test MAE while generating 453.8 KB of C code and executing in under 100 microseconds on the ESP32, which is negligible relative to the sensor’s 50 ms timing budget. These results demonstrate the practical value of multi-environment calibration for low-cost ToF mapping while also showing that environment-specific models can fail severely outside their training domain. Because the present pooled evaluation uses only the two observed environments, performance on a completely unseen environment remains an open question.

Index Terms—Time-of-Flight, LiDAR, sensor calibration, TinyML, edge AI, ESP32, indoor mapping, point cloud.

I. INTRODUCTION

Indoor 3D mapping is crucial for mobile robotics, structural inspection, and augmented reality. Commercial LiDAR systems can provide high accuracy and dense measurements, but their cost and power requirements may be unsuitable for educational platforms, low-cost robotics, and embedded sensing applications [1], [2]. A common low-cost alternative is to mount a single-point ToF sensor on a two-axis pan-tilt mechanism and convert sequential range measurements into a 3D point cloud [3].

The practical limitation of this approach is measurement reliability. Low-cost ToF sensors are affected by multipath interference (MPI), target reflectivity, ambient infrared radiation, and angular scanning geometry [4], [5]. In mechanically actuated platforms, these optical effects are further coupled with servo non-linearity and alignment error. Consequently,

a calibration model that performs well in one room may not remain accurate when lighting, wall material, or scene geometry changes.

Machine learning practically models nonlinear residual errors without requiring exhaustive analytical derivation for all optical and mechanical effects [6]. However, two issues remain important for low-cost embedded mapping. First, the calibration model should be evaluated across substantially different acquisition conditions rather than only within one controlled setting. Second, the final model must be small and fast enough to execute directly on a microcontroller.

This work addresses these issues using a VL53L1X-based pan-tilt scanner. Alongside measured distance and commanded angles, we record sensor telemetry (signal and ambient return rates) to capture measurement context. We then compare environment-specific calibration with pooled multi-environment calibration and study the accuracy-memory trade-off required for ESP32 deployment.

The main contributions of this paper are as follows:

- 1) **Cross-environment transfer analysis:** We quantify the failure of an ML calibration model when it is trained in one acquisition environment and evaluated in another with substantially different lighting, materials, and scene conditions.
- 2) **Multi-environment pooled calibration:** We show that combining data from both observed environments enables substantially lower overall calibration error than either the raw sensor or single-environment transfer models.
- 3) **Embedded TinyML deployment:** We identify a compact decision-tree model that retains most of the accuracy of the higher-capacity model while fitting the memory and latency constraints of the ESP32.

Importantly, the pooled evaluation in this study should be interpreted as multi-environment calibration over the two observed domains rather than strict unseen-domain generalization. Evaluation on an independent third environment is left for future work.

II. RELATED WORK

A. ToF Error Characterization and Multipath

The error characteristics of ToF sensors are strongly dependent on optical and geometric conditions. Dark or weakly

reflective targets reduce the number of returned photons and increase measurement uncertainty, while strong ambient illumination increases shot noise and can degrade the signal-to-background ratio [4], [7]. Measurement error can also vary with target distance, radial position, device temperature, and material properties [4], [8]. MPI introduces an additional source of bias when emitted light reaches the receiver along more than one optical path, as can occur near corners, reflective surfaces, or translucent objects.

Although much of the published metrology literature focuses on imaging ToF cameras, similar physical effects are relevant to a mechanically scanned single-point ToF sensor. In a pan-tilt scanner, optical ranging error is additionally coupled to servo angle uncertainty and mechanical alignment.

B. Machine Learning for Depth Correction

Data-driven depth correction offers an alternative to analytical calibration. DeepToF demonstrated that learned models map complex ToF distortions to corrected depth estimates [9]. Convolutional models further reduce MPI artifacts by exploiting spatial context [10], with ongoing research exploring learning-based MPI correction under diverse conditions [11].

Most high-capacity approaches target cameras or GPUs. In contrast, this work addresses a single-point ToF sensor on a low-cost scanner, emphasizing cross-environment behavior and microcontroller deployment. The goal is not algorithmic novelty, but determining whether a compact calibration model remains effective across heterogeneous conditions while executing in real time on the edge.

C. TinyML and Edge-Based Calibration

TinyML focuses on executing learned models directly on resource-constrained embedded devices [12], [13]. Although neural networks can be compressed or quantized, tree-based models are attractive for microcontrollers because inference can be represented as a sequence of conditional branches with minimal runtime dependencies. Code-generation tools such as `m2cgen` can transpile trained estimators into C-compatible source code, allowing calibration to be integrated directly into the sensor-processing loop.

For the ESP32-based platform considered here, the practical design objective is therefore not only minimum prediction error but also acceptable flash usage and inference latency.

III. METHODOLOGY

A. Hardware Platform

The sensing platform consists of a VL53L1X single-point ToF module mounted on a custom 3D-printed pan-tilt assembly actuated by two SG90 micro-servos. An ESP32 microcontroller controls the servos and communicates with the ToF sensor over I2C at 400 kHz. A DHT11 temperature sensor and an INA219 power-monitoring module were also integrated to record environmental and system variables. To ensure stable power delivery and prevent voltage drops during mechanical actuation, the servos were isolated and powered by

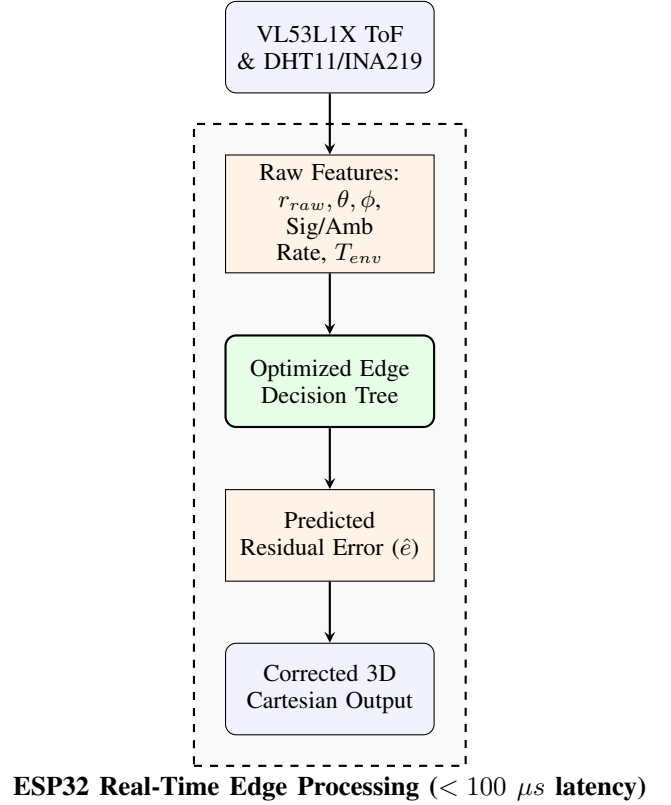


Fig. 1. System architecture and data flow. The VL53L1X provides distance and critical photon telemetry (Signal/Ambient rate) over I^2C . The ESP32 evaluates the transpiled Decision Tree to predict the residual error, applying the correction before mapping to Cartesian space.

an independent Arduino Nano. The VL53L1X timing budget was configured to 50 ms per ranging cycle.

The servo exhibits non-uniform angular error. Rather than applying mechanical correction tables, commanded angles are included directly in the feature vector, allowing the model to learn angle-dependent residuals automatically.

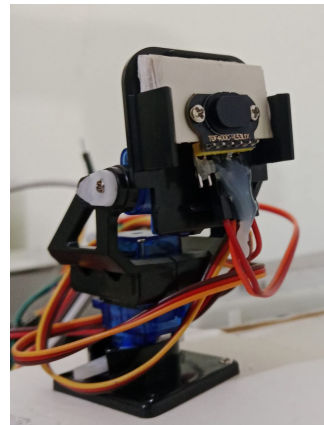


Fig. 2. Assembled pan-tilt rig wired to ESP32.

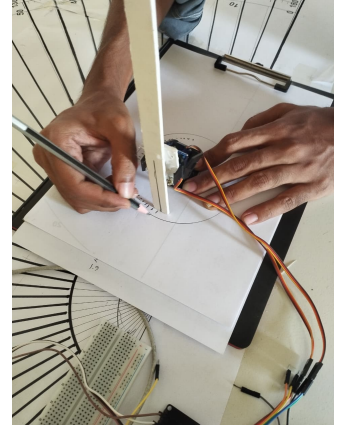


Fig. 3. Rig actively measuring servo angles.

B. Data Collection

Data were collected in two acquisition campaigns performed on separate days and in different indoor spaces. The purpose was to expose the sensor to substantially different lighting, surface, and scene conditions. The characteristics of these campaigns are summarized in Table I.

TABLE I
DATASET CHARACTERISTICS AND ACQUISITION CONDITIONS

Metric	Day 1 (Controlled A)	Day 2 (Diverse B)
Environment	Dark room, uniform walls	Hallway, LED lit, boxes
Trainable Rows	51,000	8,477
Total Valid Returns	53,539	9,104
Pan Range	0° to 180°	20° to 160°
Tilt Range (Raw)	0° to 179.9°	70° to 150°
Lighting	Low/controlled ambient	Extreme variations

In Environment A, a large reference was generated using a custom tiled protractor. Angular lines were extended toward the walls, and radial ground-truth distances were measured manually. Over 53,000 valid measurements were collected before filtering.



Fig. 4. Ground-truth setup during Day 1. A custom-printed tiled protractor defines angular reference lines on the floor, while a precise measuring tool maps the true distance to the walls.

Environment B introduced a wider range of conditions. Ten scans were recorded under varying LED illumination (partial, full, dark). Cardboard, translucent plastic, highly reflective white, and absorptive dark surfaces were introduced to trigger complex reflectance and multipath errors. Of 11,337 raw measurements, 8,477 were retained as trainable rows.

C. Residual Calibration Formulation

Instead of predicting absolute range directly, the learning problem is formulated as residual correction. Let r_{raw} denote



(a) Box with low-reflectivity wall (b) Translucent object in path

Fig. 5. Physical setup of the Environment B (Day 2) testbed. The complex scenarios feature varied lighting conditions, low reflectivity surfaces, and translucent obstacles designed to aggressively challenge the raw sensor.

the measured ToF range and r_{GT} the corresponding geometric ground truth. The residual target is:

$$e = r_{GT} - r_{raw} \quad (1)$$

The model predicts $\hat{e}$, and the corrected distance is computed as:

$$r_{corr} = r_{raw} + \hat{e} \quad (2)$$

The feature vector contains measured range, commanded pan angle θ , commanded tilt angle ϕ , signal return rate, ambient return rate, and ambient temperature. These variables were selected to represent both scan geometry and observable sensing conditions.

D. Evaluation Protocol

Three evaluation settings are considered. First, an XGBoost model is trained on Environment A and tested on Environment B. Second, the training and test environments are reversed. These two experiments quantify direct cross-environment transfer. Third, the two datasets are combined and models are trained and evaluated on the pooled dataset to measure the benefit of multi-environment calibration.

For the embedded decision-tree experiments, performance is reported on a held-out test split of the combined dataset. Specifically, we utilized a randomized 80/20 train/test split at the point level to evaluate the models on unseen data. The primary error metric is mean absolute error (MAE) in millimeters. Model footprint and measured inference latency are additionally reported for the embedded implementation.

IV. EXPERIMENTAL RESULTS

A. Cross-Environment Transfer

The two acquisition environments differed sharply in raw sensing difficulty. In Environment A, the uncalibrated sensor produced an MAE of 18.24 mm. In Environment B, which contained stronger illumination variation, different wall properties, cardboard obstacles, and translucent objects, the raw MAE increased to 934.97 mm.

This difference was also reflected in the learned calibration models. When XGBoost was trained only on Environment A and evaluated on Environment B, MAE increased to 947.15 mm. Training only on Environment B and evaluating on Environment A produced an MAE of 1049.09 mm. These

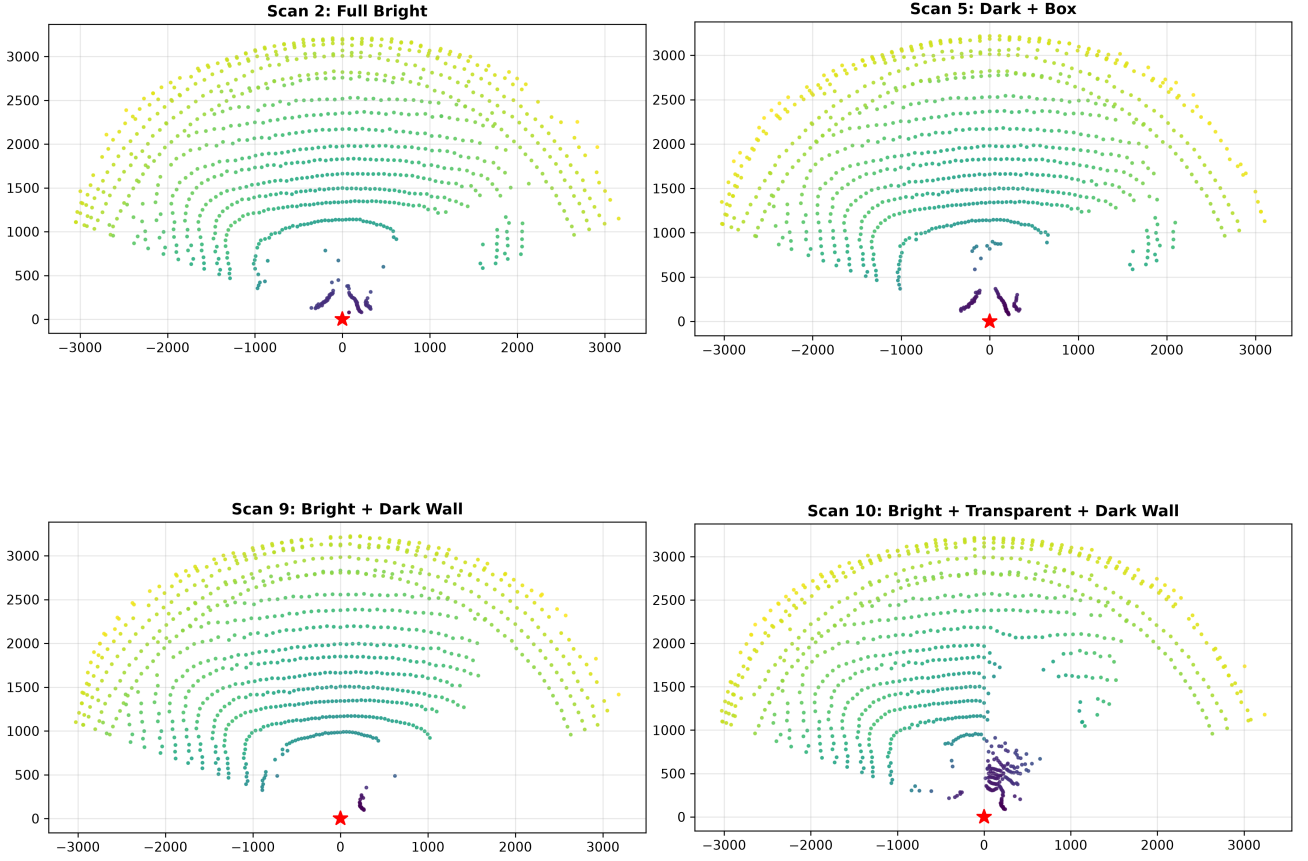


Fig. 6. Point cloud representations of four representative test scenarios from Day 2. The ML model reduces errors across challenging cases like translucent objects and varying LED illumination.

results show that a calibration model can learn corrections that are highly specific to its acquisition domain and can therefore perform poorly when transferred directly to another environment.

TABLE II
CROSS-ENVIRONMENT EVALUATION (MEAN ABSOLUTE ERROR)

Training Env.	Testing Env.	Raw MAE	XGBoost MAE
Day 1 (Controlled A)	Day 2 (Diverse B)	934.97 mm	947.15 mm
Day 2 (Diverse B)	Day 1 (Controlled A)	18.24 mm	1049.09 mm
Combined A+B	Combined A+B	149.84 mm	22.41 mm

B. Multi-Environment Pooled Calibration

When data from both environments were pooled during training, performance improved substantially across the combined dataset. The raw pooled MAE was 149.84 mm, whereas XGBoost reduced the error to 22.41 mm.

This result indicates that a single nonlinear model can represent substantially different error regimes when it is exposed to heterogeneous training data. The available features include both geometric variables and sensor telemetry, allowing the model to condition its correction on measurement context.

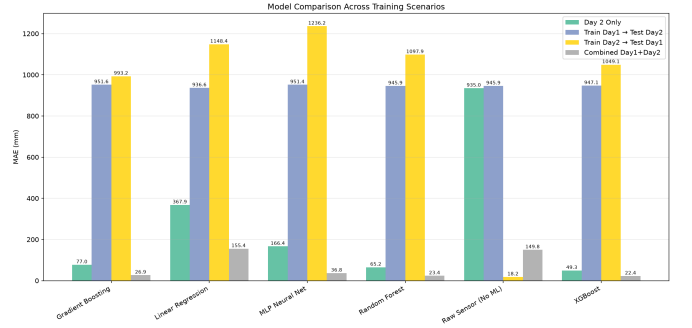


Fig. 7. Model performance comparison on the Combined Dataset. Tree-based models (RF, XGBoost, DT) significantly outperform the raw sensor and linear regression.

However, because the present experiment uses only the two observed environments, the 22.41 mm result should be interpreted as pooled multi-environment performance rather than evidence of generalization to a previously unseen room.

C. Feature Ablation and Telemetry Limitations

A logging failure corrupted the Day 1 signal-rate field (recorded as constant 15.0 kcps), and sigma was unavailable for Day 2. Sigma was therefore excluded from the model.

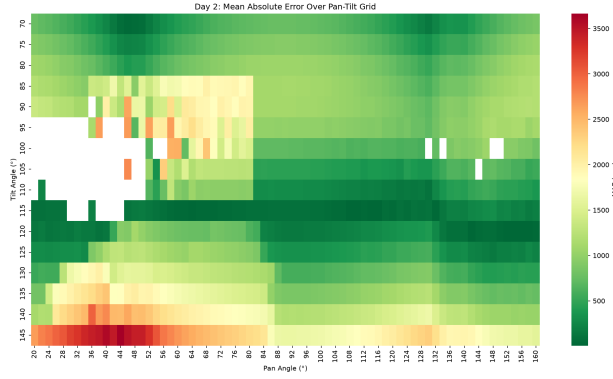


Fig. 8. Heatmap of raw residual Error (uncalibrated) across the pan-tilt scanning grid in Environment B. The ML model learns to dynamically correct these high-error regions caused by specific geometric angles and lighting.

To estimate how strongly the pooled model depends on telemetry, an ablation experiment was conducted using only range and commanded pan-tilt angles. With the full available feature set, XGBoost achieved 22.41 mm MAE. With geometry-only features, the MAE increased to 27.58 mm (Table III).

TABLE III
FEATURE ABLATION STUDY (COMBINED DATASET)

Feature Set (XGBoost)	Test MAE (mm)
All Features (Distance, Pan, Tilt, Signal, Ambient, Temp)	22.41
Geometry Only (Distance, Pan, Tilt)	27.58

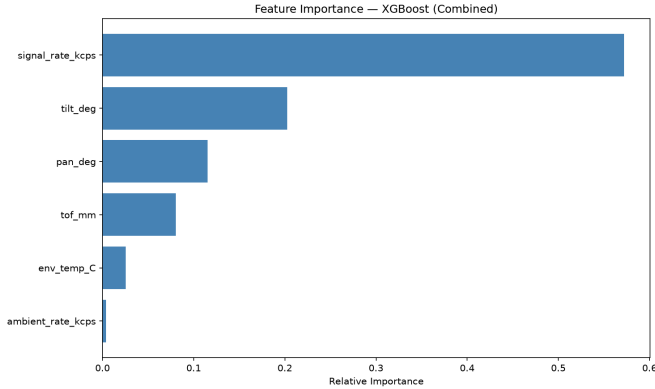


Fig. 9. XGBoost Feature Importance (Combined Dataset). While raw distance is critical, signal telemetry—especially signal rate—carries significant weight, enabling the model to dynamically adjust to different environments.

The full feature set’s improvement suggests non-geometric variables contribute useful information. However, due to the corrupted Day 1 signal-rate channel, we cannot causally claim signal rate alone drives this improvement. Future work should capture complete telemetry across all environments to evaluate individual feature groups.

D. TinyML Model Selection and ESP32 Deployment

The best pooled accuracy was obtained with XGBoost, but direct code generation produced a source representation of approximately 40 MB, which is impractical for an ESP32 with 4 MB of flash storage. We therefore performed a decision-tree depth sweep to identify a smaller model with comparable accuracy.

TABLE IV
EDGE DEPLOYMENT CHARACTERISTICS (ESP32)

Model Architecture	Test MAE	Generated Code Size	Latency
Decision Tree (Depth 10)	26.52 mm	160.2 KB	< 100 μ s
Decision Tree (Depth 12)	23.42 mm	453.8 KB	< 100 μ s
Decision Tree (Depth 15)	21.92 mm	1,212.7 KB	Exceeds target footprint

The depth-12 tree provided the optimal accuracy-memory trade-off, achieving 23.42 mm test MAE with 453.8 KB of C code. The transpiled model compiled directly onto the ESP32, with inference latency under 100 μ s—negligible relative to the sensor’s 50 ms timing budget.

E. Point-Cloud Reconstruction

The calibrated ranges were converted to Cartesian coordinates to visualize the effect of residual correction on reconstructed geometry. In the controlled environment, the raw point cloud already captured the room structure reasonably well, while calibration produced flatter planar surfaces (Fig. 10). In the more difficult second environment, the raw scans contained larger distortions and multipath-related artifacts, particularly around obstacles and translucent objects. After calibration, the reconstructed surfaces became more coherent and some large ghost points were reduced (Fig. 6).

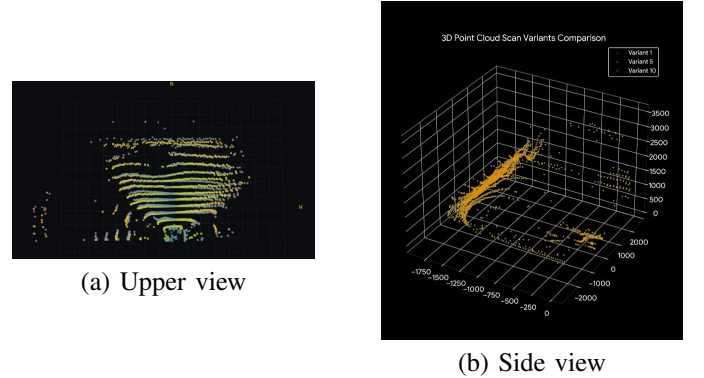


Fig. 10. Baseline 3D point cloud reconstruction of a single scan from the Day 1 controlled environment, demonstrating the sensor’s fundamental mapping capabilities from multiple perspectives.

These visualizations support the quantitative results, but they should be interpreted qualitatively. Future work should add geometric reconstruction metrics, such as plane-fitting residuals or point-to-plane error, to complement range MAE.

V. DISCUSSION

A. What the Cross-Environment Results Show

The cross-environment experiments demonstrate that calibration learned from one room can become invalid when

the sensing conditions change substantially. This finding is important for low-cost ToF mapping because the sensor error is not determined only by distance. Lighting, material reflectivity, multipath geometry, and the mechanical scan configuration all contribute to the observed residual.

The pooled result shows that exposing a model to multiple error regimes can substantially improve performance over the aggregate dataset. The result should not, however, be interpreted as proof of unseen-domain generalization. A stronger validation protocol would train on two or more environments and evaluate on an independent environment that was never used during fitting or hyperparameter selection.

B. Embedded Accuracy-Memory Trade-off

The comparison between XGBoost and the decision-tree models illustrates an important engineering trade-off. The highest-accuracy model was not the most practical model for deployment. A single depth-12 tree increased MAE only slightly relative to XGBoost while reducing the generated source footprint enough for embedded use. This makes the final calibration pipeline compatible with a low-cost microcontroller and avoids dependence on external compute or cloud inference.

C. Limitations

This study has four limitations: only two environments were tested; the Day 1 signal-rate was corrupted; the manual ground-truth procedure introduces label uncertainty; and the randomized 80/20 point-level split may overestimate generalization due to spatial correlation.

These limitations suggest clear next steps: collect a third independent environment, record complete sensor telemetry, estimate ground-truth uncertainty through repeated measurements, and evaluate using scan-level or environment-level holdout protocols. Reporting median absolute error, RMSE, upper-percentile errors, and failure-rate statistics would also provide a fuller characterization of the extreme error distribution observed in Environment B.

VI. CONCLUSION

This paper presented a low-cost pan-tilt ToF mapping platform and evaluated machine-learning-based residual calibration across two distinct indoor environments. Environment-specific models transferred poorly across the two domains, with cross-environment XGBoost MAEs near one meter. In contrast, pooled training across both observed environments reduced aggregate MAE from 149.84 mm to 22.41 mm. A memory-constrained depth-12 decision tree achieved 23.42 mm test MAE, generated 453.8 KB of C code, and executed in under 100 microseconds on an ESP32.

The results support two practical conclusions. First, calibration for low-cost ToF mapping should be trained on heterogeneous sensing conditions rather than a single controlled room. Second, a compact tree model can provide nearly the same calibration accuracy as a larger ensemble while remaining deployable on a resource-constrained microcontroller. Future

work will focus on blind evaluation in additional environments, improved ground-truth metrology, complete telemetry capture, and integration with dynamic SLAM pipelines.

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